

DATA FORT

How to Run the Project - Updated Technical Setup Guide

Personal Digital Vault | ASP.NET Core .NET 10 | Angular 21 | SQL Server

Package update: This edition adds the exact direct packages declared by the current backend *PersonalDigitalVault.API.csproj* and frontend *package.json*, with a project-specific reason for each package.

1. Project Overview

Data Fort is a full-stack Personal Digital Vault application providing authentication, email verification, password recovery, secure vault management, public document sharing, search, and administrator functions.

The supplied run guide identifies the main technology stack as ASP.NET Core .NET 10, Entity Framework Core 10.0.11, SQL Server, Angular 21.2.x, TypeScript 5.9.x, Tailwind CSS 4.1.12, JWT Bearer authentication, AES-based protected data, SHA-256 integrity hashing, and Swagger/OpenAPI.

2. Package Summary

Area	Direct packages	Role in project
Backend / API	6	Authentication, OpenAPI, EF Core, SQL Server, Swagger
Frontend runtime	9	Angular application runtime, forms, routing, async data
Frontend development	10	Build, CSS, compiler, formatting, and testing
Total	25	Direct package entries declared by the project files

Scope note: this list covers direct packages declared by the application. It does not enumerate every transitive dependency under `node_modules`.

3. Backend NuGet Packages

Package	Version	Type	Reason used in Data Fort
Microsoft.AspNetCore.Authentication.JwtBearer	10.0.11	Runtime	JWT Bearer authentication for validating access tokens and protecting secured API endpoints.
Microsoft.AspNetCore.OpenApi	10.0.11	Runtime	Adds OpenAPI support so the API can expose endpoint contracts and metadata.
Microsoft.EntityFrameworkCore.Design	10.0.11	Design-time	Provides EF Core design-time functionality for migrations and database schema development.
Microsoft.EntityFrameworkCore.SqlServer	10.0.11	Runtime	SQL Server provider that connects Entity Framework Core to PersonalDigitalVaultDB.
Microsoft.EntityFrameworkCore.Tools	10.0.11	Design-time	Provides EF Core tooling for commands such as database update and migration operations.
Swashbuckle.AspNetCore	10.2.3	Runtime / Docs	Generates Swagger/OpenAPI documentation and the interactive Swagger UI used to test the API.

4. Frontend Runtime Packages

These packages are declared under **dependencies** and are part of the Angular application's runtime.

Package	Version	Type	Reason used in Data Fort
@angular/common	^21.2.0	Runtime	Common Angular directives, pipes, and browser-oriented utilities used by the application.
@angular/compiler	^21.2.0	Runtime	Angular template compiler support required for compiling Angular templates.
@angular/core	^21.2.0	Runtime	Core Angular framework package for components, dependency injection, bootstrapping, and core APIs.
@angular/forms	^21.2.0	Runtime	Form controls and reactive-form functionality used for user input and validation.
@angular/platform-browser	^21.2.0	Runtime	Connects Angular to the browser environment and DOM.
@angular/router	^21.2.0	Runtime	Client-side SPA navigation between application views without full page reloads.
rxjs	~7.8.0	Runtime	Observable-based asynchronous programming used by Angular services and data flows.
tslib	^2.3.0	Runtime	Shared TypeScript helper functions used by compiled TypeScript output.
zone.js	^0.16.3	Runtime	Supports Angular's async/change-detection integration in the browser.

5. Frontend Development Packages

These packages are declared under **devDependencies**. They support development, compilation, CSS processing, formatting, and testing.

Package	Version	Type	Reason used in Data Fort
@angular/build	^21.2.23	Build	Angular build tooling for compiling, bundling, and preparing the frontend.
@angular/cli	^21.2.23	CLI	Angular command-line tooling including ng serve and ng build.
@angular/compiler-cli	^21.2.0	Build	Angular compiler CLI used to compile Angular applications and templates.
@tailwindcss/postcss	^4.1.12	CSS Build	Tailwind CSS v4 integration with the PostCSS processing pipeline.
jsdom	^28.0.0	Testing	Simulated DOM environment for frontend tests that require browser-like APIs.
postcss	^8.5.3	CSS Build	CSS processing layer used by the Tailwind/PostCSS build pipeline.
prettier	^3.8.1	Formatting	Consistent source formatting for readability and maintainability.
tailwindcss	^4.1.12	Styling	Utility-first CSS framework used for the application's UI styling.
typescript	~5.9.2	Language / Build	TypeScript language, compiler, type checking, and transpilation for Angular.
vitest	^4.0.8	Testing	Unit-testing framework and test runner configured for the frontend.

6. How the Major Packages Work Together

Angular core + platform-browser + common form the browser application foundation. **Angular router** handles SPA navigation, while **Angular forms** handles input and validation. **RxJS** supports asynchronous service/data flows.

JWT Bearer protects the API using access tokens. **EF Core + SQL Server provider** connect the entity model to SQL Server. **EF Core Design/Tools** support migration and database-development workflows.

OpenAPI + Swashbuckle provide API documentation and Swagger UI. **Tailwind CSS + PostCSS integration** provide the UI styling pipeline. **TypeScript + Angular compiler/build tooling** transform the frontend source into browser-ready assets.

7. Run the Project

Backend: open a terminal in PersonalDigitalVault.API

```
dotnet restore
dotnet ef database update
dotnet build
dotnet run --launch-profile https
```

Backend URL: https://localhost:7166

Swagger: https://localhost:7166/swagger

HTTP: http://localhost:5223

Frontend: open a second terminal in PersonalDigitalVault.Frontend

```
npm ci
npm start
```

Frontend: http://localhost:4200

API base URL: https://localhost:7166/api

8. Required Configuration

The current application requires secret configuration for Jwt:Key, DocumentEncryption:Key, DocumentEncryption:KeyId, Smtplib:Host, Smtplib:Port, Smtplib:User, and Smtplib>Password. .NET User Secrets is recommended for local development.

9. Database Setup

Database: PersonalDigitalVaultDB

Connection string:

Server=localhost;Database=PersonalDigitalVaultDB;Trusted_Connection=True;TrustServerCertificate=True;

EF Core migrations included: **InitialCreate**, **AddEmailVerificationFields**, and **AddEmailVerificationTokenTable**.

The startup initializer creates the User and Administrator roles but does not seed the requested login accounts.

10. Test Login Accounts

Account	Email	Password	Role
Administrator	Piriyaram2004@gmail.com	Piriyaram2004@	Administrator
Normal User	Piriyaram2025@gmail.com	Piriyaram2025@	User

The source run guide notes that these credentials are not automatically inserted by the repository. On a new database, the user records must exist with the matching roles, be active, and have IsEmailVerified = true.

11. Main Features to Verify

Authentication: registration, email verification, login, profile, password change, forgot/reset password.

Secure Vault: folders, nested folders, document upload/download, integrity verification, encrypted credentials, search.

Public Sharing: create, update, revoke, delete share links and public download by token.

Administration: dashboard, users/status, and audit logs.

12. Quick Troubleshooting

JWT error: configure Jwt:Key.

Encryption error: configure DocumentEncryption:Key and DocumentEncryption:KeyId.

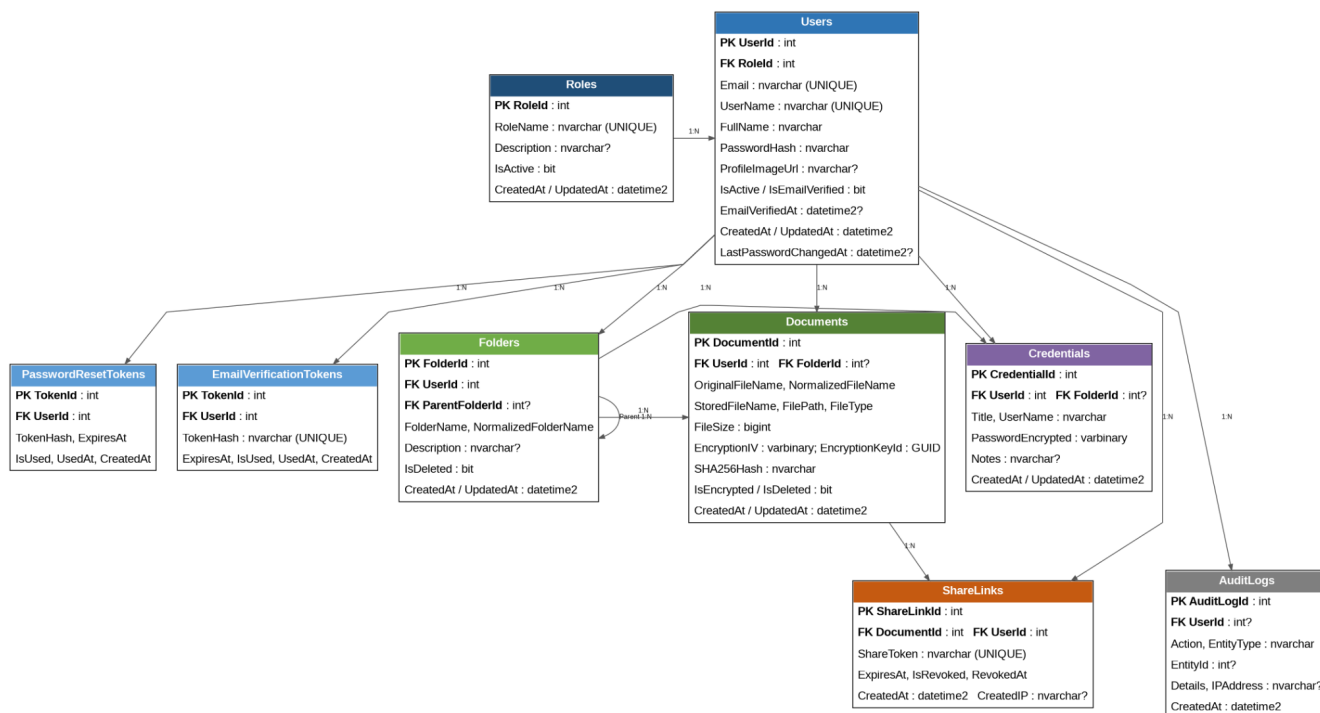
SMTP error: configure all Smtplib settings.

Angular API error: confirm https://localhost:7166 is running and the frontend uses https://localhost:7166/api.

HTTPS warning: run dotnet dev-certs https --trust.

13. Complete ER Diagram

This appendix uses the ER diagram supplied for the Data Fort project. It represents the current nine-table database model and the relationships between the entities.



Tables represented: Roles, Users, PasswordResetTokens, EmailVerificationTokens, Folders, Documents, Credentials, ShareLinks, and AuditLogs.

14. Source-Based Project Notes

The existing run guide states that the ZIP contains source code and EF Core migrations but no database backup file. It also identifies the development endpoints, secret configuration requirements, and the requested test credentials.

Prepared as an updated Data Fort run guide with the direct package inventory and project-specific reasons added.